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A review

Properties of the iodine molecule relevant to laser-induced fluorescence experiments in gas flows

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Abstract. Laser-induced fluorescence of iodine seed molecules can be used for both flow visualization and accurate measurements of gasdynamic properties. This paper gives an introductory review of the use of iodine in experimental fluid mechanics, including basic formulas for saturated and non-saturated fluorescence (excited with narrowband or broadband lasers), line shape, line strength and quenching behavior. Techniques for the seeding of the molecules into the gas flow and the safe handling of the gas are discussed. Finally, a simple numerical example is given for the calculation of absorption and fluorescence signals with discussion of the measurement of gasdynamic properties.

1 Introduction

In modern gasdynamics, there is a continuous demand for more versatile and accurate experimental techniques for flow visualization and measurements of gasdynamic properties. Laser-induced fluorescence (LIF) is among the most attractive techniques since it can provide time-resolved, 2-dimensional information about temperature, pressure, velocity and density.

Iodine is a uniquely suited seed molecule for LIF measurements. It conveniently absorbs in the visible region of the spectrum (e.g. at the green 514.5 nm line of the argon ion laser), fluoresces in a visible yellow, has reasonable values for vapor pressure, absorption cross-section and fluorescence yield, and it is only mildly toxic. Its utility for flow measurements is demonstrated.

The most attractive alternative is oxygen (Lee et al. 1986; Massey and Lemon 1984). Oxygen requires strong ultraviolet lasers (low fluorescence yield due to predissociation) and its use for measurements of gasdynamic parameters is still subject to extensive research. However, it may become the molecule of choice in the future.

A second, very useful molecule is the non-toxic biacetyl (2,3-butanedione) which has been successfully utilized by Epstein (1977) for density measurements via its fluorescence and by Hiller et al. (1984) for velocity measurements through its long-lived phosphorescence. Other flow properties appear to be difficult to extract from its luminescence due to the broad and continuous absorption spectrum.

Sodium has been used successfully for flow visualization and velocity measurements by Miles (1975) and Zimmermann and Miles (1980). However, it requires heating of the gas and experimental care due to its reactivity. NO has been used for the visualization and temperature measurements of nonreacting and reacting flows (Seitzman et al. 1985), but it is very toxic. Among the other halogens, chlorine has been used as an absorbing species in tomography by Faris and Byer (1986), but its fluorescence yield is small due to predissociation.

This list shows that iodine is a very attractive seed molecule at least in noncombusting gas flows (iodine is not chemically stabe in a reactive environment, another reason for the interest in oxygen). In addition to its use in flow visualization by Teshima (1983) and McDaniel and Graves (1986), iodine has beeen utilized to measure pressure (McDaniel 1984; Hiller and Hanson 1988), temperature (Fletcher and McDaniel 1987), density (McDaniel 1982, 1983), and velocity (McDaniel 1983; Hiller et al. 1983; Hiller and Hanson 1985, 1988) both in single points of the flow field and simultaneously at multiple points in a plane.

The motivation for this paper is to review the properties of iodine and to provide a resource for anyone who wishes to use laser-induced fluorescence of iodine in experimental gas dynamics. The experimental section (Sect. 3) of this paper is written based on our experience with a small wind tunnel (length scales on the order of centimeters) but it should provide a good starting point for readers with larger facilities in mind. The spectroscopy described is universal.

After an introduction to the basic principles of laser-induced fluorescence, a simplified set of formulas will be listed which enables the calculation of the signal intensities that can be expected in a given flow situation (Sect. 2). Experimental techniques for the seeding of the iodine molecules to the flow as well as the safe handling will be discussed (Sect. 3). Section 4 will focus on the spectroscopy of iodine

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and will provide the information necessary to evaluate some of the parameters that appear in the equations. Finally, a typical numerical example will be given in Sect. 5.

2 Absorption/fluorescence model

A similar setup is employed in all laser-induced fluorescence schemes (Fig. 1). Gas molecules absorb a portion of the incident laser radiation, and a fraction of the absorbed radiation is subsequently emitted as fluorescence in all directions. By detecting the fluorescence in a direction which is perpendicular to the exciting laser beam, a high depth resolution is achieved which is only limited by the thickness of the laser beam. While the physical process is the modulation of the absorption by concentration, temperature, pressure or velocity, its effect is detected by collecting the emitted fluorescence. The lateral resolution is limited by the collection optics and the size of the detector.

In comparison with the other molecular light scattering techniques based on the Raman effect or molecular Rayleigh scattering (Exton and Hillard 1986; Long et al. 1985), laser-induced fluorescence is a much more efficient process so that the signal levels are orders of magnitude higher. The laser beam can be generously expanded into a wide sheet to illuminate an entire plane. This plane can then be imaged onto an array of detectors to provide two-dimensional information about the flowfield. By sweeping the illuminating plane with a rotating mirror, quasi-3D information can be obtained (Yip and Long 1986). The sheet of radiation can be formed either by expanding the laser beam with a cylindrical telescope, or by reflecting the beam back and forth between two mirrors (forming a so-called multipass cell). By means of imaging optics, the laser-induced fluorescence in the flow is imaged onto the detector array, so that each picture element (pixel) of the photodiode-array camera receives fluorescence from a well-defined volume.

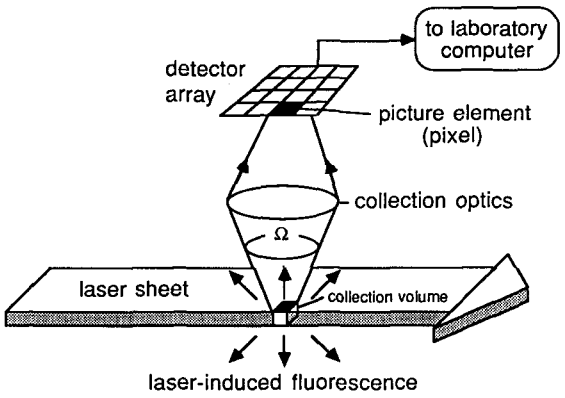

Fig. 1. Basic imaging setup for laser-induced fluorescence schemes

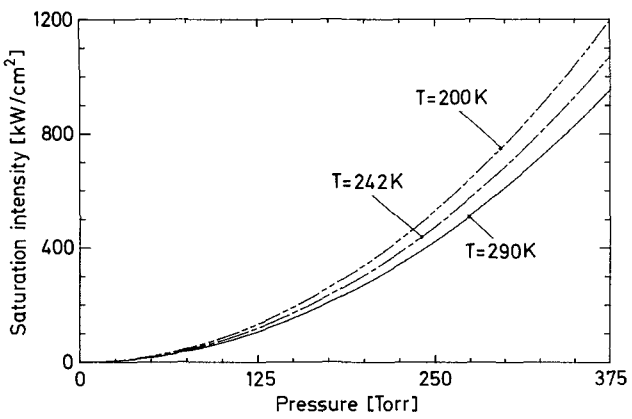

Fig. 2. Saturation of the P13/R15 (43, 0) *B-X* transition of iodine vs. static gas pressure, with the static gas temperature as parameter; the saturation intensity is calculated for narrow-band excitation at line center

For iodine, the absorption/fluorescence mechanism can be described with a simple two-level model as was shown by McDaniel (1982) in his pioneering work on the use of iodine fluorescence for quantitative measurements of gas properties. In what follows, the important formulas for iodine absorption and fluorescence will be briefly listed (Hiller 1986).

According to Beer's law, the absorbed power P_a in volume V is proportional to the size of the volume and the incident laser intensity I:

$$P = \alpha V I . \tag{1}$$

The proportionality factor is the so-called absorption coefficient α:

$$\alpha = \frac{\alpha_0}{1 + \frac{I}{I_s}} . \tag{2}$$

α_0 is he unsaturated absorption coefficient and describes the basic susceptibility of the molecules to absorption. As the laser intensity approaches the saturation intensity I_s, more and more molecules are in an excited energy state (after having absorbed photons) and a larger fraction of the photons passes through the volume unabsorbed. The net effect is a decrease of the absorption coefficient. The saturation intensity is physically reached when there is one photon with energy $h\nu$ incident on each atom, within its absorption cross section σ, per recovery time τ:

$$I_s = \frac{h\nu}{\sigma\tau} . \tag{3}$$

This simple argument neglects stimulated emission, but is a good first-order approximation which permits the judgement whether saturation has to be considered. Figure 2 shows how the saturation intensity varies with pressure and temperature. The calculations underlying Fig. 2 include stimulated emission. h is Planck's constant, ν is the optical frequency of the exciting photon

$$\nu = \frac{c}{\lambda} . \tag{4}$$

Here, c is the speed of light in vacuum and λ is the wavelength in vacuum. The recovery time τ is inversely proportional to the sum of all rates by which the absorbed energy is removed from the molecule:

$$\tau = \frac{1}{A_{21}+Q} . \tag{5}$$

A_{21} is the Einstein coefficient of spontaneous emission (=fluorescence) and Q is the quenching rate (removal of the absorbed energy by collisions with other gas molecules). The latter competes with the fluorescence and is desired to be as small as possible for strong signals. Both quantities will be described in Sect. 4. The fluorescence efficiency η_f, to be used later, can now be introduced. It is described as the fraction of excited molecules which undergo spontaneous emission (i.e. emit fluorescence):

$$\eta_f = \frac{A_{21}}{A_{21}+Q} . \tag{6}$$

The unsaturated absorption coefficient α_0 is the product of the absorption cross-section σ and the number density of the absorbing molecules.

$$\alpha_0 = \sigma\, n_a . \tag{7}$$

The absorption cross-section can be expressed as

$$\sigma = h\nu \frac{B_{12}}{c} \left\{ \begin{array}{ll} \times g & \text{for narrowband excitation} \\ \times \dfrac{1}{\Delta\nu_l} & \text{for broadband excitation} \end{array} \right\} . \tag{8}$$

B_{21} is the Einstein coefficient of absorption (related to the coefficient of spontaneous emission A_{21}, Sect. 4.2), and g is the absorption line shape function (Sect. 4.4). Narrowband excitation means that the laser linewidth is much smaller than the absorption line width [i.e. when a single transverse (or axial) laser mode is selected with an intracavity etalon]. In the broadband case, the laser spectral linewidth $\Delta\nu_l$ is much wider than the absorption line (in the absence of an intracavity etalon or with a pulsed laser). The number of absorbing molecules n_a is proportional to the total number density of molecules n, the mole fraction of iodine molecules χ_a and the fractional population f_p of iodine molecules in the quantum states which are able to absorb at this particular laser wavelength (Sect. 4.1):

$$n_a = n\, \chi_a\, f_p . \tag{9}$$

With the equations listed so far, the absorbed power P_a in the control volume can be readily computed. Of the absorbed power, only a fraction η_f is emitted as fluorescence, and of this emitted radiation, only a fraction η_c is collected with the imaging optics. η_c is called collection efficiency and can be calculated from

$$\eta_c = \frac{\Omega}{4\pi} = \left[\frac{1}{4f/\#}\left(\frac{M}{M+1}\right)\right]^2 . \tag{10}$$

Here, Ω is the solid collection angle, $f/\#$ the f-number of the lens (ratio of focal length to lens diameter) and M the imaging magnification (ratio of image height to object height, or ratio of image distance to object distance).

The fluorescent power P_f incident on the detector is then simply:

$$P_f = \eta_c\, \eta_f\, P_a . \tag{11}$$

With these simple expressions and the extension given in Sect. 4, the fluorescence signal can be estimated with reasonable accuracy.

One important effect of saturation can be seen if one looks at the limit of $I \gg I_s$. The absorbed power is then independent of the laser intensity [Eqs. (1) and (2)]:

$$P_a = \alpha_0\, V\, I_s ; \quad (I \gg I_s) . \tag{12}$$

Moreover, the power of the fluorescent radiation is independent of the quenching rate (which is often difficult to determine with a reasonable accuracy), as shown by Eqs. (3)–(7), (11) and (12),

$$P_f = \eta_c\, h\nu\, A_{21}\, V\, n_a . \tag{13}$$

The fluorescence is thus directly proportional to the number density of absorbing molecules, and hence the total number density if χ_a is fixed, independent of the exciting laser linewidth or intensity. Of course, the mole fraction of iodine molecules must be known and the absorbing ground state has to be chosen such that it is insensitive to temperature changes in the region of interest [Eq. (16)].

Saturated laser-induced fluorescence is thus very attractive for the measurement of number density. A major problem which has limited the use of saturated fluorescence methods is that laser power variations across the beam result in a spatially varying saturation in the flowfield. This disadvantage can be partially offset by using 'flat' beam profiles or other clever geometric schemes (Cottereau 1986).

If number density is to be measured with a nonsaturating narrowband laser, the signal is greatly influenced by the other thermodynamic properties, through the quenching and lineshape function terms. Detuning the laser far into the wing of the absorption line appears to be the only viable solution (McDaniel 1982, 1983), although the signal strength suffers. Often it is best to calculate density based on temperature and pressure measurements.

The other gasdynamic quantities of interest can be measured as follows: pressure via the line shape function g, temperature via the fractional population f_p and velocity via the Doppler shift of the line shape with respect to the laser optical frequency (see references in the introduction and brief discussion at end of Sect. 5).

3 Experimental techniques

This section addresses the practical aspects of the use of iodine as a seed molecule: seeding technique, collection of

the optical signal, the protection of the facility from iodine corrosion, and health effects. As stated in the introduction, our experience is based on a small wind tunnel with length scales on the order of centimeters. This tunnel was operated in an open loop with a vacuum pump providing the pressure difference. The air was exhausted through a char coal filter. An extension to a larger facility may be possible, but this facility would have to be run in a closed loop to recycle the iodine gas.

3.1 Seeding technique

The iodine is seeded to the gas flow by simple sublimation from iodine crystals (e.g. Baker 4-2208) and forms a stable diatomic molecule in the gas phase at the conditions of interest. The carrier gas should not contain water vapor for two reasons. Iodine gas reacts with water to form iodic acid (HI) which attacks metals faster than iodine itself. Secondly, the iodine crystals appear to form a surface coating under the influence of water which inhibits the sublimation process and leads to reduced fluorescence signals.

The choice of the carrier gas is an optimization between quenching behavior (Sect. 4.3), effect on the speed of sound through the ratios of the specific heats (relevant for transonic and supersonic applications), velocity slip, toxicity, reactivity and price. In consideration of these factors, nitrogen was chosen as the carrier gas in our experiments, although oxygen, dry air, argon are also suitable. It was shown experimentally by Teshima (1983) and theoretically by Hiller (1986) that velocity slip is not a problem in nitrogen-based flows up to moderate Mach numbers (<5) where the velocity is equilibrated within about 10 collisions.

In our facility, the entire nitrogen flow was sent through the mixing chamber containing the iodine crystals. The mixing-chamber should be isolated with inlet and outlet valves, not only to contain the iondine safely during down time, but also to enable the adjustment of the mixing chamber pressure to select the iodine seeding fraction. Assuming saturation of the carrier gas, the seeding fraction can be estimated from the ratio of the iodine vapor pressure at reservoir temperature (0.3 Torr, Fig. 3) and the mixing chamber pressure indicated by a pressure gauge. For larger facilities, temperature control to a few degrees via a heating tape may be necessary. This value establishes the upper limit. Absorption measurements proved that seeding fractions within 10% of the saturation value were achieved regularly.

A plot of the range in which the iodine mole fraction is sufficiently low to avoid significant absorption over a 1 cm long path is given in Fig. 4. Since absorption scales linearly with distance in the unsaturated case, the acceptable mole fraction can be easily estimated for different path lengths. Let's assume, for example, that the mixing chamber pressure is adjusted to atmospheric pressure (750 Torr) which, according to Fig. 4, corresponds to a iodine mole fraction of 400 ppm. If temperature and pressure of the gas flow of

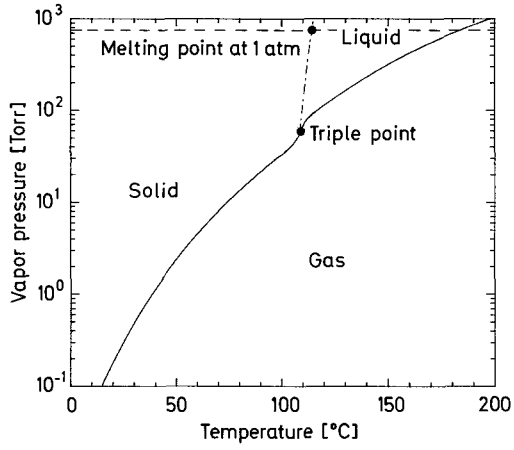

Fig. 3. Phase diagram of iodine (solid line describes the vapor pressure)

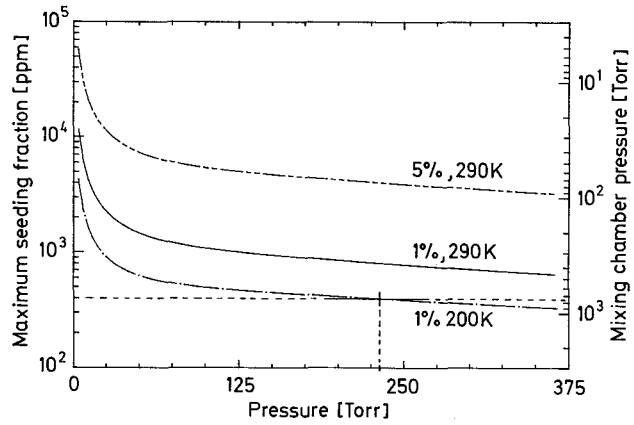

Fig. 4. Iodine seeding curves with allowed absorption (over 1 cm length) and gas flow temperature as parameters [narrow-bandwidth excitation at line center of P13/R15 (43, 0) *B-X* transition]; the mixing chamber is assumed to be at room temperature; the cross refers to the example given in the text

interest are 200 K and 232 Torr, respectively, then the absorption along the laser beam path in through the flow will be 1%/cm (Fig. 4). For quantitative measurements, the absorption across the entire flow field should not exceed 1%. A smaller mole fraction may be chosen if the signal strength remains adequate. This can be desirable in larger facilities.

For flow visualization purposes where optical thickness is not a concern, the largest mixing chamber pressure is desirable and heating tape can be wrapped around the mixing chamber to further enhance the iodine mole fraction. Since the wavelength of the fluorescence is red-shifted from the laser wavelength (Sect. 3.2), scattered light from iodine crystals in the gas flow is easily blocked with a long-pass filter. Therefore, from a light scattering viewpoint, particles in the flow are acceptable as long as they do not lead to optical thickness (i.e. laser beam intensity loss). In applications with high-power pulsed lasers, fluorescence of the iodine particles is possible, and any particulate contamination can pose a

more serious problem that needs to be solved with suitable filters.

For optimum signal strength and control of the seeding fraction via the mixing chamber pressure, saturation of the carrier gas with iodine is desired. This requires a large surface area of the crystals and a sufficiently long residence time in the mixing chamber. The best solution appears to be a fluidized-bed mixing chamber with a cross-section that increases toward the top to slow down the flow and sufficient height to ensure that the crystals fall back and are not carried into the flow chamber (cyclone-type flow).

Many techniques to measure flow properties rely on relative measurements (Sect. 5) so that uniform seeding is not required and wall losses are not too critical if a reasonable signal strength can be maintained. If uniformity is a concern (for instance in 3-D visualization) baffle plates and stirring devices may have to be used to mix the flow. Diffusion helps, iodine diffuses in nitrogen at about 3 mm/s at STP (Table 4). Unfortunately, we have no information on the rate of loss of iodine to walls. In mixing studies it is generally desirable to inject the iodine-laden flow into a pure main flow (McDaniel and Graves 1986).

3.2 *Fluorescence detection*

As indicated in Fig. 1, the emitted fluorescence is detected at 90° with a single detector for point measurements or a detector array for linear or planar measurements. A commercial lens (in our case a Nikon 50 mm $f/1.4$ lens) can be used to image the collection volume onto the front of the detector. The lateral extent of the collection volume is determined by the size of the detector and the resolution of the imaging arrangement. In general, geometric optics equations are fully adequate to determine the lateral extent of the collection volume. Focussing errors can degrade the spatial resolution significantly if low-f/number lenses are used. The depth of the collection volume is determined by the thickness of the laser beam (or sheet), usually measured at the $1/e^2$ intensity points.

Iodine fluorescence, when excited with an argon ion laser, appears yellow to the eye. A close look at the fluorescence spectrum (sketched in Fig. 5) reveals a regular structure. One can identify the sharp spikes of the primary vibrational spectrum, caused by the radiative decay from the excited molecular energy level to all possible ground states, and the significantly weaker vibrational transfer spectrum: collisions between molecules transfer the excited iodine molecules to adjacent energy levels, from which they can also decay to lower levels. Since most of these collisions result in quenching of the fluorescence, the vibrational transfer spectrum is much weaker than the original spectrum. Figure 5 shows that the strongest fluorescence can be expected from a decay to the original ground level. This kind of fluorescence is termed resonance fluorescence since it is in resonance with the optical frequency of the exciting laser radiation.

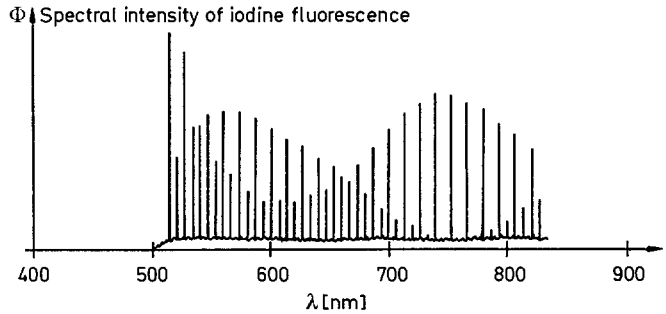

Fig. 5. Sketch of the fluorescence spectrum of iodine when excited with the green 514.5 nm line of an argon ion laser

Unfortunately, elastically scattered radiation from surfaces or particles interferes with this resonance fluorescence and produces an undesirably large and nonuniform background noise. It is, therefore, desirable to block all elastically scattered radiation (including the resonance fluorescence) and to detect only the red-shifted fluorescent decay to higher energy levels than the ground state. The potential to employ spectral filtering is an inherent advantage of nonresonant laser-induced fluorescence techniques and allows the complete elimination of elastic background scattering.

Spectral filtering is best done with an optical long-pass filter which only passes radiation above a certain wavelength. We chose a 560 nm longpass filter while using the 514.5 nm line of our argon ion laser as the excitation source. If the material of the longpass filter is absorbing glass (which is the least expensive kind), luminescence of the glass under background radiation is possible but can be circumvented by using an additional laser notch filter. The filters should be placed directly on the detector and all potentially reflecting surfaces in the facility should be black.

3.3 *Protection of the facility*

The iodine molecule is very reactive. It forms a powdery corrosive surface on metals such as aluminum, copper and steel. After the initial coating, further corrosion is sufficiently slow to eliminate concerns about leaks on a time scale of several years. Uncoated brass fittings can be used where surface quality is not important. The surface has to be coated if smoothness is required, e.g. in the nozzle. Flat black paint (Ferrothane no. 31, Flecto, Inc.) has proven to be effective, although black-anodization of aluminum surface (i.e. coating with an aluminum oxide layer) results in a superior finish. Stainless steel reacts only weakly with iodine, but forms a powdery surface. Glass, teflon and PVC do not react, but are eventually coated with a layer of condensed iodine which can be easily removed with ethanol. The flow system should contain teflon tubing and stainless steel valves. The sealing of the facility can be accomplished with O-rings and iodine resistant fluorocarbon vacuum grease (Fisher Flurolube grease GR-90).

3.4 Health effects

Iodine is harmful when inhaled. Irritations of eyes, nose, mouth (sore throat) and the respiratory tract occur. The maximum accepted concentration is 0.1 ppm, which corresponds to 1 mg/m^3 STP (Bruin 1980). Fortunately, the detection limit of the receptors in the nose lies below this limit. The enclosure of all iodine vapors in the experiment is absolutely necessary. Reported lethal doses in adults range from a few tenths of a gram to more than 20 g. Probably, the mean lethal dose lies between 2 and 4 g (Gosselin et al. 1976). The most effective treatment of chronic iodine overdoses (resulting e.g. in sneezing, skin rashes or bronchitis) is discontinuation of the exposure. After discontinuation, no long-term effects of chronic overdoses have been observed (Gosselin et al. 1976).

4 Spectroscopy of the iodine molecule

To evaluate the relevant factors which appear in Eqs. (1)–(13) (the fractional population f_p, the Einstein coefficients A_{21} and B_{12}, the quenching rate Q, and the absorption line shape g), a good understanding of the spectroscopy of the iodine molecule is necessary.

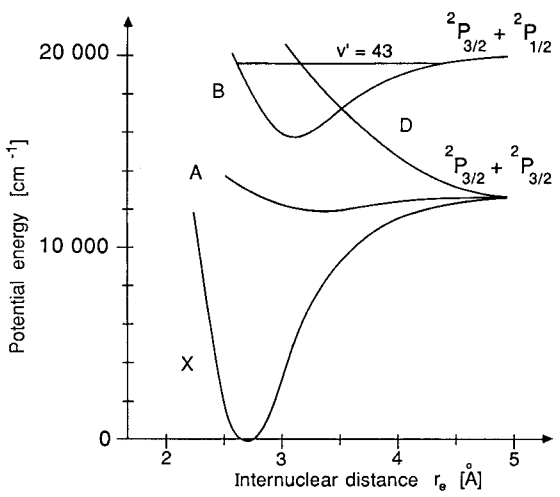

Fig. 6. Potential energy curves of the iodine molecule

The electronic transition which produces the part of the visible spectrum used with visible LIF occurs from the ground electronic X state, a singlet, to the B state which is a triplet. The B-X transition is between states of different electronic spin and is therefore only weakly allowed. This is the reason for the long natural lifetimes (between 0.3 and 7 µs; Capelle and Broida 1973), compared with typical lifetimes in the visible part of the spectrum of 10 ns. Figure 6 shows the potential energy curves of the bound X, A and B states and the unbound D state. The molecule predissociates into $2P_{1/2}+P_{3/2}$ atoms for excitation energies above 20,037 cm^{-1} (below 499 nm). The spectroscopic data tabulated in Table 1 are collected from Herzberg (1950) and Huber and Herzberg (1979).

Absorption can occur between rovibrational states if the selection rule $\Delta J = \pm 1$ is observed. J is the rotational quantum number. P branches ($\Delta J = -1$) and R branches ($\Delta J = +1$) can exist. About 150 rotational levels and 3 vibrational levels are populated at room temperature. Together with about 50 addressable upper vibrational levels, this results in some 45,000 visible absorption lines of the B-X electronic band in the region between 500 and 650 nm. This close spacing is a result of the large mass of the iodine atom (127 amu) which causes a small eigen-frequency of vibration and a large moment of inertia in the molecule. A good picture of the iodine absorption spectrum is provided by figures taken from Capelle and Broida (1973) and Gerstenkorn and Luc (1978), in connection with a plot of the lines around the argon-ion gain profile (see also Table 2), and is displayed in Fig. 7. The line positions are taken from Gerstenkorn and Luc (1978), the assignments from Koffend et al. (1979) and Rapagnani and Davis (1979). The transitions of main interest are the overlapping P13/R15 lines of the (43,0) band at 514.5 nm under the gain profile of the strong green line of the argon ion laser.

Iodine absorption lines exhibit hyperfine splitting due to the interaction of nuclear and molecular spin (Levinson and Schawlow 1972). A total of 36 hyperfine lines are observed for the P13/R15 lines. Rotational lines with even values of J are therefore split into 15 hyperfine components, states with

Table 1. Molecular constants of the iodine molecule

State Assignment			X $^1\Sigma_g^+$	A $^3\Pi_{1u}^+$	B $^3\Pi_{0u}^+$	D $^1\Sigma_u^+$
Electronic energy	T_e	[cm^{-1}]	0	11,888	15,769	33,744
Electronic temperature	θ_e	[K]	0	17,150	22,750	48,993
Vibrational constant	w_e	[cm^{-1}]	214.5	44.0	125.7	104.0
Vibrational temperature	θ_v	[K]	309.5	63.5	184.6	150.0
Unharmonic vibrational constant	$w_e x_e$	[cm^{-1}]	0.613	1.0	0.773	0.2
Rotational constant	B_e	[cm^{-1}]	0.03735		0.029	
Rotational temperature	θ_r	[K]	0.054		0.0421	
Vibrational-rotational coupling	α_e	[cm^{-1}]	1.13×10^{-14}		1.58×10^{-14}	
Internuclear distance	r_e	[Å]	2.666		3.016	

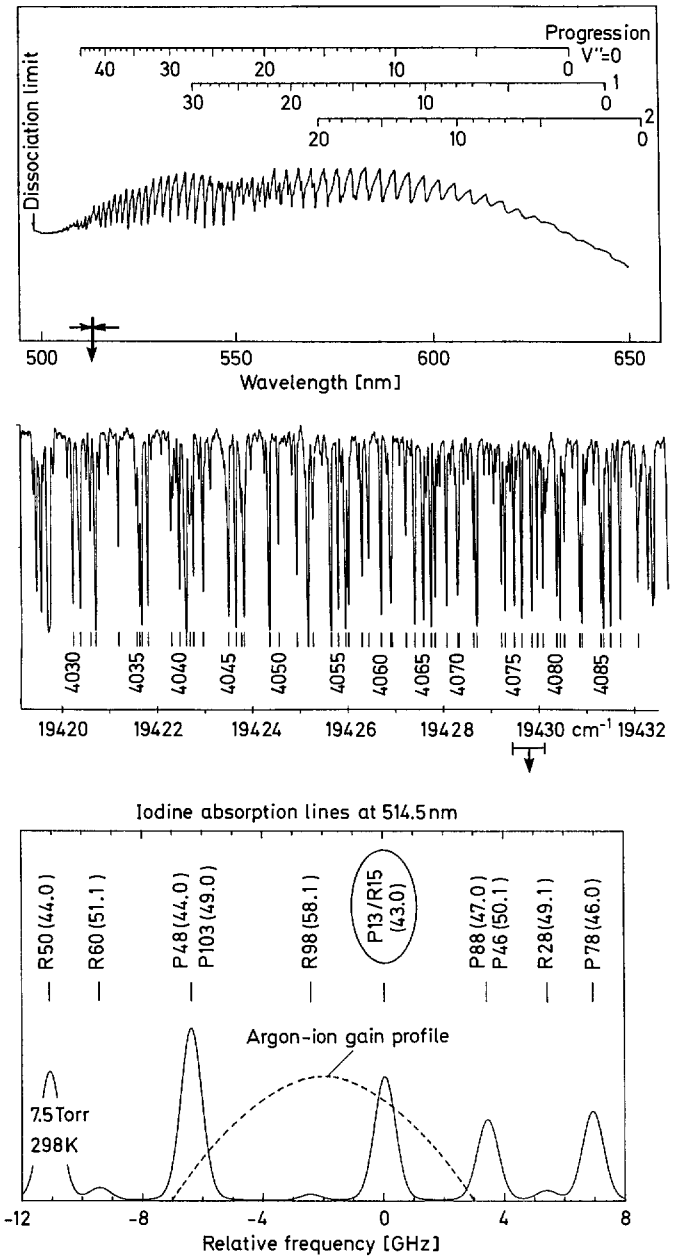

Fig. 7. Absorption spectrum of the *B-X* system of iodine, showing a two-step zoom-in process on the iodine transition which is located under the green 514.5 nm argon ion laser line

odd J into 21 hyperfine components. These components cannot be resolved at normal flow conditions and were investigated with Doppler-free spectroscopy. This splitting leads to a widening of the homogeneous line width and has implications for the fractional population (Sect. 4.1).

4.1 Fractional population

With Boltzmann's law, the fractional population f_p in a particular energy state ε can be written as (Vincenti and Kruger 1965):

$$f_p \simeq C_1 (2J+1) \frac{\theta_r}{T} (1-e^{-\theta_v/T})\, e^{-\varepsilon/kT}. \tag{14}$$

The constant C_1 takes the value 5/6 for even J and 7/6 for odd J (resulting from the fact, that both even and odd J's are allowed, and that the nuclear spin moment of iodine is 5/2; Hiller 1986). Rotational temperature θ_r and vibrational temperature θ_v are listed in Table 1. The term kT can be easily computed from $kT\,[\mathrm{cm}^{-1}] = 0.693\,T\,[\mathrm{K}]$.

The energy level of the absorbing state ε can be expressed using the terms in Table 1:

$$\varepsilon = T_e + w_e\left(v+\frac{1}{2}\right) - w_e x_e \left(v+\frac{1}{2}\right)^2 + B_e J(J+1) - \alpha_e\left(v+\frac{1}{2}\right) J(J+1). \tag{15}$$

v is the vibrational quantum number.

With $v''=0$ (a single prime denotes the upper state and a double prime the lower state) and neglecting the weak vibrational-rotational coupling, the level with the smallest sensitivity to temperature changes becomes approximately

$$J = 0.83 \sqrt{\frac{T\,[\mathrm{K}]}{B_e\,[\mathrm{cm}^{-1}]}}. \tag{16}$$

Typical values for the combined population in the absorbing rotational levels 13 and 15 ($v''=0$) are 0.77% at 293 K, 1.35% at 200 K and 3.1% at 100 K. For reasonable signal strengths, a transition from $v''=0$ is desirable.

4.2 Line strength

The coefficients for stimulated transitions B_{ij} can be related to the coefficients for spontaneous transitions A_{ij} by considering the equilibrium between a gas and a blackbody radiation field. The transformation into other units is discussed by Lucht et al. (1978) and Hilborn (1982), the latter also gives a

Table 2. Line assignments, line positions relative to the center of the 514.5 nm argon-ion gain profile and Franck-Condon factors (FCF)

(J', J'')	Branch	(v', v'')	FCF	Position [GHz]
(51, 50)	R50	(44, 0)	0.01617	−9.11
(61, 60)	R60	(51, 1)	0.00416	−7.45
(47, 48)	P48	(44, 0)	0.01618	−4.42
(102, 103)	P103	(49, 0)	0.01167	−4.42
(99, 98)	R98	(58, 1)	0.00439	−0.47
(12, 13)	P13	(43, 0)	0.01779	+2.00
(16, 15)	R15	(43, 0)	0.01779	+2.00
(87, 88)	P88	(47, 0)	0.01290	+5.41
(45, 46)	P46	(50, 1)	0.00440	+5.41
(29, 28)	R28	(49, 1)	0.00470	+7.36
(77, 78)	P78	(46, 0)	0.01382	+8.90

good general description. The relationship between B and A coefficients is

$$B_{ij}\left[\frac{\mathrm{m}^3}{\mathrm{J\,s}^2}\right] = 6\times 10^4\, A_{ij}\,[\mathrm{s}^{-1}]\,\lambda^3\,[\mathrm{nm}]\,. \tag{17}$$

Since the degeneracies for the B-X system are 1, the indices i and j can be reversed. Capelle and Broida (1973) give a natural lifetime of 0.44 μs for the 43-0 vibrational band. The Einstein A coefficient is the inverse of this lifetime. With Eq. (17) we can calculate B_{12} to be $4\times 10^{18}\ \mathrm{m}^3\,\mathrm{J}^{-1}\,\mathrm{s}^{-2}$. Values for other transitions can be scaled linearly using Franck-Condon factors (Tellinghuisen 1978); FCFs for the transition under the argon ion gain profile are given in Table 2).

4.3 Quenching cross-section

If a single carrier gas is used and if the iodine mole fraction is kept small to ensure optically thin behavior (i.e. a collision with another iodine molecule is much less probable than with a carrier gas molecule), the quenching rate Q can be written as (Vincenti and Kruger 1965)

$$Q = \sigma_{q,\mathrm{carrier}}\sqrt{\frac{8\,k\,T}{\pi}}\; n\sqrt{\frac{1}{m_{I_2}}+\frac{1}{m_{\mathrm{carrier}}}}\,. \tag{18}$$

Quenching cross-sections of iodine strongly depend not only on the collision partner but also on the upper energy state and are given for the 43-0 vibrational band in Table 3 (Capelle and Broida 1973; Nicolai and Heaven 1986; Nakagawa et al. 1986).

4.4 Line shape function

In this section, the line shape function g will be analytically described and values for the significant parameters will be given. If the line shape is mainly influenced by collisional processes in the gas and by the Brownian motion of the gas molecules, which is true for the gasdynamic conditions of interest, it can be described with the Voigt function $V(x, y)$:

$$g = \sqrt{1+\frac{I}{I_s}}\sqrt{\frac{4\ln 2}{\pi}}\,\frac{1}{\Delta\nu_i}\,V(x,y)\,. \tag{19}$$

I_s is the saturation intensity from above, I the laser intensity. The other parameters are defined below. The Voigt function constitutes the real part of the complex probability function, which can be looked up for a given x and y in mathematical tables, e.g. the "Handbook of mathematical functions" (Abramowitz and Stegun 1964; Table 7.9; p. 325). A very useful computer routine to compute the Voigt function is listed by Humlicek (1979). With the relationships for the complex probability function, it is also possible to compute the derivatives of the Voigt function, for example with respect to optical frequency or pressure (Hiller 1986).

The two variables are the detuning parameter x,

$$x = 2\sqrt{\ln 2}\,\frac{\Delta\nu}{\Delta\nu_i} \tag{20}$$

and the broadening parameter y:

$$y = \sqrt{\ln 2}\,\frac{\Delta\nu_h}{\Delta\nu_i}\sqrt{1+\frac{I}{I_s}}\,. \tag{21}$$

The detuning $\Delta\nu$ can be written in terms of the laser frequency ν, the line center (= resonance) frequency ν_r and the line shift with respect to a certain reference pressure $\Delta\nu_p$:

$$\Delta\nu = \nu - \nu_r - \Delta\nu_p\,. \tag{22}$$

The pressure shift at a certain pressure P and temperature T from the reference conditions P_0 and T_0 was measured by McDaniel (U. Virginia, personal communication 1986) to:

$$\Delta\nu_p = -1.4\ \mathrm{MHz}\left(\frac{P\,[\mathrm{Torr}]}{\sqrt{T\,[\mathrm{K}]/298}}-\frac{P_0\,[\mathrm{Torr}]}{\sqrt{T_0\,[\mathrm{K}]/298}}\right). \tag{23}$$

The homogeneous linewidth ν_h for the transition used is (McDaniel 1982)

$$\Delta\nu_h = 12\ \mathrm{MHz}\,\frac{P\,[\mathrm{Torr}]}{\sqrt{T\,[\mathrm{K}]/298}}\,. \tag{24}$$

The inhomogeneous linewidth $\Delta\nu_i$ is, to a good approximation, the sum of the width due to hyperfine splitting (333.5 MHz; Levenson and Schawlow 1972) and the Doppler width

$$\Delta\nu_D = 210\ \mathrm{GHz}\,\frac{1}{\lambda\,[\mathrm{nm}]}\sqrt{\frac{T\,[\mathrm{K}]}{m\,[\mathrm{amu}]}}\,. \tag{25}$$

where m is the molecular mass. With this information, the line shape function can be computed for a wide range of temperatures and pressures. Broadening parameters for other iodine transitions are not known to the authors.

Table 3. Collision cross sections and quenching cross sections (room temperature) of iodine [Å^2]

Gas:	H_2	He	N_2	O_2	Ar	CO_2	Xe	I_2
Quenching	0.63	0.79	3.00	3.24	5.11	7.40	10.6	65.0
Collision		4.6	5.9		5.5			7.8

4.5 Collision cross-section

For gaskinetic calculations, it is often important to know the collision cross section. Therefore, an estimate of the collision cross section of iodine with several collision partners is listed in Table 3 for completeness. It is based on a simple model of elastic spheres and values from "Chemical engineer's handbook" (Perry et al. 1973).

Table 4. Typical numbers (calculated for the overlapping P13/R15 lines of the 43-0 vibrational band of the *X-B* electronic system of iodine, at 125 Torr and 298 K)

Spontaneous, natural lifetime	2	μs
Radiative recovery time	16	ns
Fluorescence efficiency	0.8%	
Absorption linestrength B_{12}	4×10^{18}	$m^3/J\,s^2$
Franck-Condon factor	0.01779	
Homogeneous (collisional) linewidth	1,500	MHz
Doppler width	442	MHz
Hyperfine width	333.5	MHz
Inhomogeneous linewidth	775.5	MHz
Voigt broadening parameter y	1.6	
Voigt function at line center (Including influence of adjacent lines)	0.63	
Fractional population in absorbing state	0.77%	
Absorption cross-section	3.5×10^{-16}	cm^2
Saturation intensity	100	kW/cm^2
Diffusion coefficient (nitrogen, STP)	0.07	cm^2/s
Molecular mass of iodine	254	amu
Absorbed photon energy $h\nu$	3.86×10^{-19}	J

5 Example

The following numerical example will illustrate how the material in this paper can be used to calculate the fluorescence intensity that can be expected from a given flow situation and experimental arrangement. Many of the numbers calculated here are also listed in Table. 4.

A typical medium-size argon ion laser has an output power of 4 W. At least 1 W can be expected in single axial mode operation of the green 514.5 nm line. With a beam diameter of 1 mm, measured at the $1/e^2$ intensity points, the average intensity is 130 W/cm^2. Let us assume that the laser is tuned to center of the overlapping P13/R15 absorption lines (either by temperature or angle tuning of the intracavity etalon). Figure 6 shows that the laser power will decrease to 75% (or 100 W/cm^2) of the peak value because the center of the laser line and the center of the absorption lines do not coincide.

Let us further assume that the laser beam travels through a static cell with iodine gas at 125 Torr and 298 K and a path length of 1 cm. What fluorescence signal can we expect if a section of the beam path of 1 mm is imaged onto our detector?

With any fluorescence measurement, two essential checks should be made first: (1) Will the laser intensity saturate the absorption? In that case the absorption line will broaden [Eq. (21)] and the signal will no longer be linear with laser intensity [Eq. (2)] so that quantitative results may be difficult to obtain. (2) Will there be significant absorption across the beam path? The fluorescence signal will then depend on the flow conditions along the laser beam prior to reaching the observation point. Quantitative fluorescence results can be only obtained if there is no significant absorption, i.e. the medium is optically thin.

From Fig. 2, the saturation intensity at the conditions set above is 100 kW/cm^2, three orders of magnitude higher than the actual laser intensity. Hence, saturation is not a problem in this case, $I/I_s \simeq 0$ and $\alpha = \alpha_0$. However, if the beam is tightly focussed to achieve high spatial resolution (e.g. to a diameter of 30 μm), saturation is easily achieved.

To calculate the optical thickness, the pressure of the mixing chamber containing the iodine crystals needs to be known. For safety reasons, the mixing chamber is best kept at 1 atmosphere to minimize gas exchange with the outside through possible leaks. From Fig. 4 we read an iodine mole fraction χ_a of 450 ppm, for which the absorption along the 1 cm pass is significantly less than 1%. Hence, the medium is optically thin.

The next step is to calculate the power absorbed in a 1 mm long section of the beam path (defining a collection volume of 0.8 mm^2) with Eq. (1). This requires the evaluation of the absorption cross section σ and the number density of absorbing molecules n_a. [The product of both is the absorption coefficient α Eq. (7).]

For the absorption cross section, we need the line strength value (more precisely, the Einstein coefficient of absorption) B_{12} (Sect. 4.2). This value is based on a Franck-Condon factor of 0.01779 (Table 2). Section 4.4 shows how to compute the line shape function. The value x for the Voigt function (Eq. (20)] is 0 since we are tuned to the center of the absorption line. Value y [Eq. (21)] is 1.6. The Voigt function is then $V(x, y) = 0.306$. If we consider that we are effectively dealing with two lines and if we include line strengths, line positions and fractional populations of adjacent lines, the values for the line shape function g at line center is 0.76/GHz and the absorption cross section becomes 3.5×10^{-16} cm^2. [The product of line shape, line strength and fractional population are simply summed up, taking in consideration the different relative line positions – the Voigt value x is only 0 for the two overlapping lines but needs to be evaluated with Eq. (20) for all adjacent lines!]

The total number density (computed from the gas equation $n = P/k\,T$, k being Boltzmann's constant) is in this case 4×10^{18} cm^{-3}. Only 450 ppm of these molecules are iodine molecules. And only a fraction f_p of these molecules is in a ground energy state that can be excited to resonance with the single-mode argon laser. This fractional population can be computed with Sect. 4.1 to be 0.77%. The number density of absorbers is then 1.4×10^{13} cm^3.

With these values, the absorption coefficient is about 5×10^{-3} cm^{-1} or 0.5%/cm (supporting the estimate above using Fig. 4). The absorbed power in the control volume is then approximately 0.5 mW. To calculate the power of the fluorescence, we need to evaluate the fluorescence efficiency and the optical collection efficiency.

The fluorescence efficiency is calculated with Eqs. (6) and (18) to be 0.8%. With an imaging magnification of $M = 2$ and an $f/1.4$ lens, the collection efficiency η_c is 1.4%. The use of a long-pass filter in combination with a laser notch filter (recommended in Sect. 3.2) will reduce this value to approx-

imately 1%. The fluorescence power is then 40 nW and can be easily detected with a photodiode. Even at an exposure time of 1 μs, about 120,000 'yellow' iodine photons could be collected. With a quantum efficiency of the detector of 50%, the shot-noise limited signal-to-noise ratio is an impressive 240.

Focussing of the laser beam will not increase the fluorescence power if the length of the collection volume remains the same (the loss of area exactly balances the increase in laser intensity). Near saturation, focussing will result in a loss of signal.

For imaging experiments with detector arrays, the typical size of a collection volume is much smaller (by about a factor of 100) so that the time resolution and/or SNR will have to be compromised. The use of image intensifiers to amplify the fluorescence photon flux to improve time resolution in the shot noise limit may be beneficial.

With the values calculated above, we can now confirm that we are not saturating the transition. The recovery time τ is expressed as function of the radiative rate A_{21} and the quenching rate Q in Eq. (3). At the pressure of 125 Torr, the quenching rate is dominating and results in a recovery time of 16 ns (Table 4; Sect. 4.3). This recovery time and not the natural lifetime limits the time resolution in flow experiments. The photon energy $h\nu$ can be also read from Table 4. With the absorption cross section calculated above, we find a saturation intensity $I = 70\ \mathrm{kW/cm^2}$, in reasonable agreement with the more precise calculations which produced the plot in Fig. 2.

How will the fluorescence signal behave as we raise the pressure to one atmosphere? The saturation intensity will increase (Fig. 2) since there are now more absorbing molecules in the beam path so that saturation is even less a problem. The optical thickness is larger now but still below 1%, as an extrapolation of the curves in Fig. 4 would show.

The fluorescence signal will change in the following way. Quenching will increase at the same rate as the number density increases – both linearly in this situation, so that the net effect is zero. The only net difference will be introduced by the line shape function. As the pressure increases, the line broadens and the influence of adjacent absorption lines becomes larger. A calculation of the line shape function has to take this effect into account. Computer computations indicate a reduction of the line center value by approximately a factor of 3 as the pressure is raised from 125 Torr to 750 Torr. That is, the fluorescence signal itself will decrease by only a factor of 3 which indicates that iodine fluorescence should still be useful for flow visualization purposes at atmosphere pressures. However, the absorption lines then start to significantly overlap and accurate measurements of gasdynamic properties become more difficult.

In what follows we will discuss how this calculated signal relates to the measurement of flow properties. The shot-noise limit SNR was better than 200 in our example so that the precision, barring all other influences, should be better than 1%.

Temperature is best measured by taking the ratio of line center measurements at two different lines. By taking ratios, number density, seeding fraction and quenching cancel out if the two signals are collected in the same volume in quick succession. This scheme should still leave a precision of 1% but other systematic errors and experimental reality should limit the quality to an error of about 5%. Fletcher and McDaniel (1987) achieved an accuracy of 2%.

Velocity can be measured by tuning the laser over the entire line and by measuring the Doppler shift between flow and a static cell (McDaniel 1983; Hiller and Hanson 1988). If the tuning is done fast enough so that density and quenching are 'frozen', a relative measurement is again sufficient. A better, faster way is to fix the laser wavelength in the wing of the line and measure signal changes due to the Doppler shift. Several self-calibrating schemes were demonstrated by Hiller and Hanson (1988). Although the precision is similar to above, the remaining error due to the influence of the non-linear line shape and other non-ideal behaviors is about 5%.

Pressure can be also measured with above fixed-wavelength scheme with a similar accuracy. McDaniel (1984) suggests a method based on the inverse line center fluorescence value, but here the seeding fraction has to be constant. Density is best calculated from pressure and temperature measurement, or should be attempted to be measured with a saturating broadband laser.

The photon flux from other spectroscopic methods (Raman, Rayleigh) is orders of magnitude lower. Laser-induced fluorescence is, therefore, also the best candidate for measurements in turbulent flows. To 'freeze' the flow, nanosecond-scale measurements are necessary which can be only accomplished with pulsed lasers. At these short illumination times, saturation will be necessary to achieve reasonable signal levels. The measurement of gasdynamic properties in turbulent flows is still very much in its infancy and most work to date has been done in flow visualization and saturated density measurements.

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